

Project Title

Bank Transaction Analysis and Customer Behavior Insights

Project Abstract

This mini project analyses a small bank transactions dataset to explore customer financial behavior, transaction patterns, and basic anomaly detection. Using transaction-level data (deposits, withdrawals, transfers, and payments) with metadata such as payment method, location, and running balance, the project will demonstrate exploratory data analysis (EDA), feature engineering, simple visualization, and elementary predictive or rule-based detection of unusual activity. The work aims to produce summary statistics, customer-level transaction profiles, and recommendations for next steps (larger dataset modelling, fraud detection, or churn prediction).

Dataset Details

Dataset name: Bank_Transactions_Sample_Aug2026 (suggested)

Number of features/columns: 8

Transaction ID

Customer ID

Date

Transaction Type

Amount

Payment Method

Location

Balance

Target variable, if applicable:

Not strictly required for descriptive analysis. Possible targets for supervised tasks:

Fraudulent transaction (requires labels, not present here)

Customer churn (would need customer status and more history)

High-risk transaction flag (could be derived with rules)

For this phase, the project will be unsupervised/exploratory; later phases could define a target after data enrichment.

Database Details

Database name/type: SQLite or PostgreSQL (recommended for small projects). For prototyping, CSV files or a Pandas DataFrame are acceptable.

Tables to be used:

transactions (main table containing the 8 columns listed)

(optional) customers (if you add customer metadata later: name, age, city, account_open_date)

Important fields/columns:

Transaction ID: unique identifier for each record.

Customer ID: key to aggregate and analyze per-customer behavior.

Date: transaction timestamp for time-series and recency analysis.

Transaction Type: categorical (Deposit, Withdrawal, Transfer, Payment) used for segmentation.

Amount: numeric, primary measure of transaction value.

Payment Method: categorical (UPI, ATM, NEFT, IMPS, Debit Card) useful for channel analysis.

Location: branch/city where transaction occurred (Chennai, Coimbatore, Madurai, Trichy).

Balance: running account balance (useful for validating transactions and trend analysis).

Technology Stack

Python version: 3.10+ (suggested)

Libraries/frameworks:

Data handling: pandas, numpy

Visualization: matplotlib, seaborn, plotly (optional interactive)

Date/time: datetime, pandas datetime utilities

Database connectivity: sqlalchemy (for SQLite/Postgres), psycopg2 (Postgres)

Modeling / analysis: scikit-learn (for later phases), statsmodels (optional)

Database:

CSV files for initial work; SQLite for local DB; PostgreSQL for production-like setup

Development tools:

IDE: VS Code or JupyterLab/Notebook

Version control: git/GitHub

Environment: virtualenv or conda

Expected Outcome

Deliverables for Phase 1:

Cleaned dataset (CSV or DB) with consistent types and parsed dates.

Exploratory Data Analysis report including:

Summary statistics (count, mean, median, min, max) for Amount and Balance.

Transaction counts by type, payment method, and location.

Time-based view (transactions over dates) and per-customer transaction frequency.

Customer profiles: total deposits, withdrawals, net flow per customer.

Visualizations: bar charts for transaction types, heatmap or table for locations vs. transaction type, time-series plot of total daily amounts.

Basic anomaly rules and findings (examples below).

Recommendations and next steps (feature additions, possible targets, modeling ideas).

Example anomaly rules (illustrative):

Transactions where Amount exceeds 3x the customer's average transaction amount.

Sudden large withdrawals that drop the balance by >50% in one transaction.

Transactions from a location not usually associated with the customer.

Success criteria:

Clear EDA with visuals and write-up.

Reproducible notebook (Jupyter) that ingests the dataset, performs cleaning, EDA, and outputs findings.

At least 3 actionable insights or hypotheses for further modeling.

Would you like me to:

Produce the Jupyter notebook code for the EDA and visualizations using the sample data you provided?

Create CSV/SQL import statements and schema definitions?

Expand this Phase 1 into a Phase 2 plan with specific modeling approaches?